

Multimodal Sensor Fusion for Human Activity Recognition:

Exploring Data-Level and Decision-Level Fusion across Inertial Modalities on Adapted MHEALTH

Satya Siddhartha Balanagu

Student ID: A1961325

School of Computer Science, University of Adelaide

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Abstract

Wearable human activity recognition relies on body-worn inertial sensors to classify physical behaviours for remote health monitoring. Single-sensor placements suffer from anatomical blind spots where distinct activities produce similar local kinematic signatures. This study investigates multimodal sensor fusion across acceleration and angular velocity modalities using the adapted MHEALTH dataset (343,195 samples across 10 participants). An audit identifies 615,648 missing values (7.80%), where 90.4% occur as isolated single-sample gaps, consistent with a Missing Completely at Random mechanism resolved via segment-bounded interpolation. Under a subject-independent evaluation (Subjects 1–7 training, Subject 8 validation, Subjects 9–10 testing), we benchmark three unimodal multi-layer perceptrons against feature-level early fusion and decision-level late soft voting. Late fusion achieves 95.56% test accuracy and 0.9568 Macro-F1, outperforming the strongest unimodal baseline (chest acceleration, 0.8848 F1) by 7.20 percentage points and removing major confusions between walking and stair climbing, though static sitting (0.685 F1) and standing (0.807 F1) remain challenging. On-node feature extraction is estimated to reduce transmission payload by 86.2% relative to raw telemetry, supported by an ethical governance framework.

1 Task 1: Data Exploration

1.1 Dataset Provenance and Architecture

The Mobile HEALTH (MHEALTH) dataset, introduced by Baños et al. [1, 2] and hosted on the UCI Machine Learning Repository [3], provides multimodal motion and physiological recordings for human activity benchmark analysis. Ten healthy volunteers of diverse physical profile performed 12 physical activities in an out-of-laboratory setting.

Data was recorded using three Shimmer2 wearable inertial units attached to the chest, right wrist, and left ankle at a sampling frequency of $f_s = 50$ Hz ($\Delta t = 20$ ms),

which satisfies Nyquist sampling criteria for voluntary movement (< 20 Hz) [4]:

1. **Chest Sensor:** Measures trunk tilt, gravitational alignment, and respiratory/cardiac motion via a tri-axial accelerometer (m/s^2) and 2-lead electrocardiogram (ECG, mV).
2. **Left-Ankle Sensor:** Captures heel-strike impacts, propulsion, and stride frequency via a tri-axial accelerometer (m/s^2), tri-axial gyroscope (dimensionless sensor units), and tri-axial magnetometer [5].
3. **Right-Wrist Sensor:** Captures forearm rotational rate, arm swing, and gestures via tri-axial accelerometer, gyroscope, and magnetometer.

1.2 Dataset Properties and Curation

An inspection of the dataset identifies three structural properties:

1. **Activity Class Scope:** In the benchmark recordings, unlabelled transition intervals between exercises accounted for over 70% of raw samples. This work focuses on the 12 target physical activities (L_1 to L_{12}).
2. **Sample Distributions:** The dataset contains 343,195 rows across 10 subject files (mHealth_subject1.log to mHealth_subject10.log). Timed activities (L_1 – L_5 , L_9 – L_{11}) contain exactly 30,720 samples across participants (≈ 61.4 s per subject at 50 Hz). Repetition-based tasks show cadence variation (L_6 : 28,315; L_7 : 29,441; L_8 : 29,337). Jumping (L_{12}) contains 10,342 samples (≈ 20.7 s per subject), reflecting brief, explosive execution (Table 1).
3. **Sequence Permutation:** Activity execution order varies across participants (e.g., Subject 1 completes L_5 last, whereas Subject 8 swaps L_2 and L_3). Windowing and interpolation must therefore be bounded within individual activity segments to prevent cross-condition contamination.

Table 1: Taxonomy of activities, nominal protocol, movement intensity classification, raw sample volume, and segmented 2.56-second window counts ($W = 128$ samples) in the adapted MHEALTH dataset ($N = 5,229$ windows across 10 participants).

ID	Activity Name	Nominal Protocol	Movement Intensity Classification	Raw Samples	Windows
L_1	Standing still	1 min timed	Static Postural Balance	30,720	470
L_2	Sitting and relaxing	1 min timed	Static Sedentary Posture	30,720	470
L_3	Lying down	1 min timed	Static Recumbent Posture	30,720	470
L_4	Walking	1 min timed	Rhythmic Ambulatory Locomotion	30,720	470
L_5	Climbing stairs	1 min timed	Inclined Ambulatory Stepping	30,720	470
L_6	Waist bends forward	20 repetitions	Trunk Flexion / Calisthenic	28,315	429
L_7	Frontal arm elevation	20 repetitions	Upper-Limb Gesture Dynamics	29,441	446
L_8	Knees bending (crouching)	20 repetitions	Lower-Limb Crouch Calisthenic	29,337	444
L_9	Cycling	1 min timed	Rotary Lower-Limb Cadence	30,720	470
L_{10}	Jogging	1 min timed	Vigorous Dynamic Locomotion	30,720	470
L_{11}	Running	1 min timed	High-Intensity Aerobic Locomotion	30,720	470
L_{12}	Jump front & back	20 repetitions	Explosive High-Impact Burst	10,342	150
Total Structured Dataset				343,195	5,229

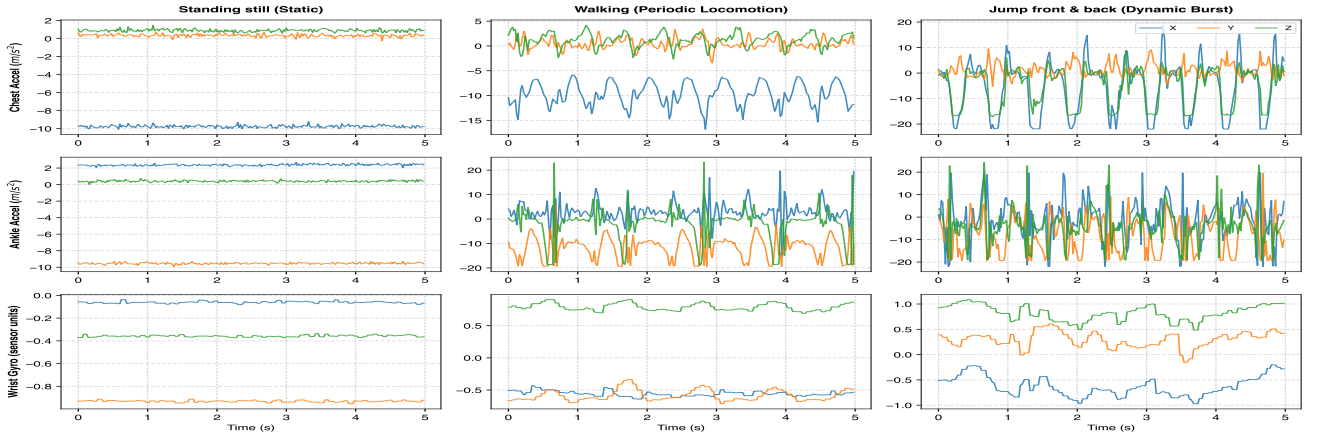


Figure 1: Representative 5-second tri-axial sensor waveform traces ($\Delta t = 20$ ms, $f_s = 50$ Hz) across contrasting biomechanical activities: Static Standing (L_1 , left), Periodic Walking (L_4 , centre), and Explosive Jumping (L_{12} , right) across Chest Acceleration, Ankle Acceleration, and Wrist Gyroscope (sensor units).

1.3 Exploratory Data Analysis: Kinematics and Synergy

We examine multi-axis kinematics across activity regimes to characterise pattern separability and sensor synergy.

1.3.1 Signal Dynamics Across Activity Regimes

Figure 1 contrasts raw sensor waveforms across three physical regimes:

- **Static Posture (Standing, L_1):** Signals show stationary profiles dominated by gravity. The Chest Accelerometer records $\approx -9.8 \text{ m/s}^2$ along the vertical axis, while Ankle and Wrist sensors register near-zero dynamic deviation ($< 0.05 \text{ m/s}^2$) and low angular rates ($\text{VM} \approx 1.0$).
- **Periodic Locomotion (Walking, L_4):** Displays sinusoidal patterns at a gait frequency of $\approx 1.8 \text{ Hz}$. Ankle acceleration exhibits dynamic swings ($[-15, +25] \text{ m/s}^2$) from cyclic heel-strike impacts, while the Wrist Gyroscope captures smooth periodic oscillations from arm swing.
- **Explosive Impact (Jumping, L_{12}):** Features in-

termittent landing spikes ($> 30 \text{ m/s}^2$) on the ankle, followed by aerial micro-gravity phases ($\approx 0 \text{ m/s}^2$) and sharp wrist rotations.

1.3.2 Movement Intensity and Inter-Subject Variance

We compute instantaneous Vector Magnitude (VM) and Signal Magnitude Area (SMA) for each window:

$$\text{VM}(t) = \sqrt{x(t)^2 + y(t)^2 + z(t)^2} \quad (1)$$

$$\text{SMA} = \frac{1}{W} \sum_{t=1}^W \sum_k |s_k(t)| \quad (2)$$

As shown in Figure 2, ankle acceleration magnitude separates physical intensity tiers, whereas wrist angular velocity magnitude maintains a narrow median across all activities (≈ 1.0 – 1.2 sensor units) and serves instead through directional and temporal dynamics. Across participant medians, ankle VM delineates:

1. **Sedentary and Low-Intensity Tiers (L_1 – L_3 , L_6 – L_8):** Ankle acceleration reflects gravitational equilibrium, with median values between 9.1 and 9.5 m/s^2 .

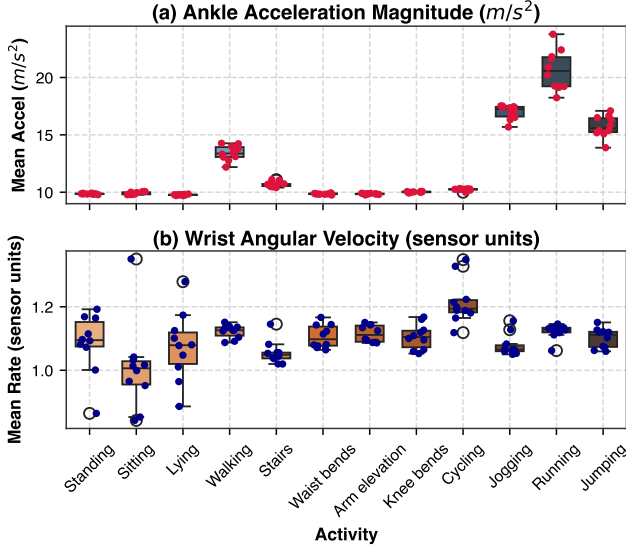


Figure 2: Movement intensity distributions across 10 participants: (a) Mean Ankle Acceleration Magnitude (m/s^2); (b) Mean Wrist Angular Velocity (sensor units). Points denote participant averages, illustrating inter-subject variation.

2. **Transitional and Moderate Ambulation** (L_5, L_9, L_4): Stair climbing (L_5) averages 10.0 m/s^2 and cycling (L_9) averages 9.7 m/s^2 (remaining low due to seated foot rotation), while walking (L_4) reaches 12.6 m/s^2 from heel impacts.
3. **Vigorous and Explosive Regimes** (L_{12}, L_{10}, L_{11}): Jumping (L_{12}) produces 14.9 m/s^2 , jogging (L_{10}) reaches 16.3 m/s^2 , and running (L_{11}) peaks at 19.7 m/s^2 .

The stripplot points in Figure 2 highlight substantial inter-subject variability; for example, during running (L_{11}), Subject 3 exhibits 31% higher ankle acceleration than Subject 4 (22.6 m/s^2 vs 17.3 m/s^2), reflecting personal cadence and foot-strike mechanics.

1.3.3 Cross-Modality Correlation and Complementarity

Figure 3 presents the Pearson correlation matrix across 17 sensor channels ($N = 50,000$). Ankle acceleration axes are weakly correlated ($r = -0.08$ X-Z, 0.06 X-Y, 0.15 Y-Z), indicating independent spatial axes. The strongest channel correlations occur between Ankle-GyrX and Wrist-GyrZ ($r = 0.60$) and between ECG Leads 1 and 2 ($r = 0.52$). Cross-modality relationships demonstrate partial complementarity:

- Wrist Gyroscope channels exhibit correlations up to $|r| = 0.50$ with chest and ankle accelerometers (e.g., Ankle-AccZ vs Wrist-GyrY), capturing coupled limb movements rather than complete statistical independence.
- ECG Leads 1 and 2 show near-zero linear correlation ($|r| \leq 0.07$) with all motion channels, operating on distinct physiological time scales.

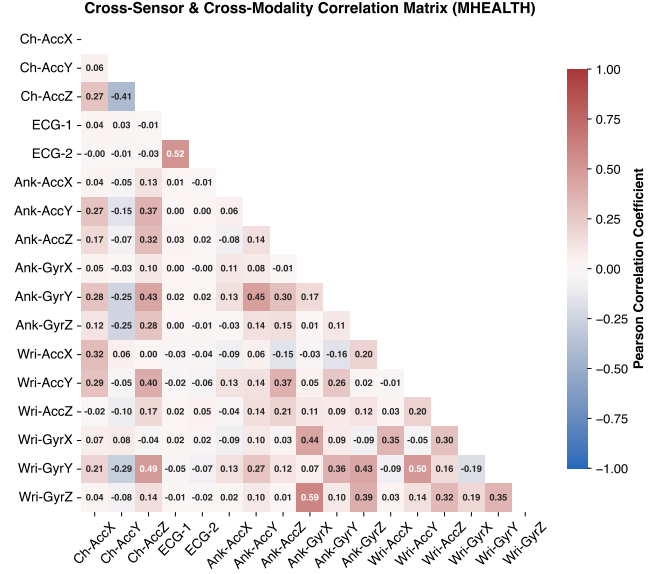


Figure 3: Cross-sensor Pearson correlation matrix across 17 kinematic and physiological channels ($N = 50,000$).

This partial redundancy indicates that combining these modalities expands decision boundaries and mitigates unimodal blind spots [6].

2 Task 2: Missing Data and Pre-processing

2.1 Quantification and Diagnosis of Missingness

Missing entries in wearable sensor streams arise from transmission dropouts, buffer overruns, or recording faults. Missing entries in the raw records are represented as IEEE NaN values.

An audit reveals **615,648 missing values** across 343,195 time steps, yielding an overall missingness rate of **7.80%** across the 23 sensor channels (Table 2). The activity label column contains zero missing values. Missingness varies across channels (Figure 4(a)), from 1.13% (Wrist Gyro-Y) to 14.67% (Ankle Mag-Z), but remains highly uniform across participants (7.75% to 7.84%) and across activities (7.72% to 7.84%).

2.1.1 Burst Length Analysis and Missingness Mechanism

To diagnose the missingness mechanism, we analyse consecutive missing sample runs (burst length $\ell = t_{\text{end}} - t_{\text{start}} + 1$). Figure 4(b) shows that **90.4% of bursts consist of a single sample** ($\Delta t = 20 \text{ ms}$). Bursts of $\ell = 2$ samples comprise 8.5%, with a maximum burst length of $\ell_{\text{max}} = 7$ samples (140 ms) across the entire dataset. The mean burst length is 1.11 ± 0.35 samples.

The occurrence of missing entries is independent of signal magnitude and participant activity ($P(\text{missing} |$

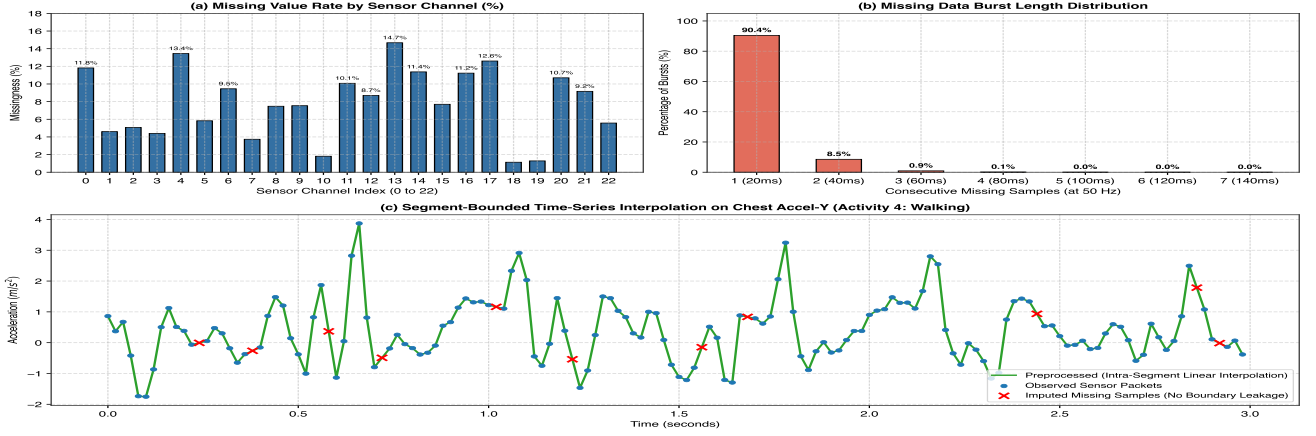


Figure 4: Missing data audit and preprocessing pipeline: (a) Percentage missingness per sensor channel (Channels 0–22); (b) Burst length frequency distribution, confirming 90.4% single-sample drops (20 ms); (c) Intra-segment linear interpolation on Chest Accel-Y during Walking (L_4), demonstrating accurate trajectory reconstruction without cross-activity boundary leakage.

Table 2: Missing data audit per anatomical sensor cluster and burst characteristics before and after preprocessing.

Sensor Source	Cols	Missing	Rate	Max ℓ	Post NaNs
Chest Accel (X, Y, Z)	0–2	73,759	7.16%	5	0
Chest ECG (Leads 1, 2)	3–4	61,252	8.92%	6	0
Ankle Accel (X, Y, Z)	5–7	65,274	6.34%	5	0
Ankle Gyro (X, Y, Z)	8–10	57,746	5.61%	4	0
Ankle Mag (X, Y, Z)	11–13	114,789	11.15%	7	0
Wrist Accel (X, Y, Z)	14–16	103,983	10.10%	6	0
Wrist Gyro (X, Y, Z)	17–19	51,543	5.01%	6	0
Wrist Mag (X, Y, Z)	20–22	87,302	8.48%	5	0
Entire Dataset	0–22	615,648	7.80%	7	0

$x, y) = P(\text{missing})$), with isolated single-sample losses dominating (90.4%). This pattern conforms to the **Missing Completely at Random (MCAR)** mechanism [7].

2.2 Segment-Bounded Intra-Run Interpolation

Because voluntary human movement has mechanical bandwidth < 20 Hz, kinematic trajectories remain smooth over sub-150 ms intervals. **Local linear interpolation** provides an accurate reconstruction:

$$\hat{s}(t) = s(t_a) + \frac{t - t_a}{t_b - t_a} (s(t_b) - s(t_a)), \quad t_a < t < t_b \quad (3)$$

where t_a and t_b are the nearest observed samples.

Preventing Boundary Leakage: Interpolating across raw file boundaries would mix adjacent activities (e.g., interpolating jumping impacts into stair climbing). To avoid boundary mixing, our pipeline enforces **segment-bounded intra-run grouping**:

1. Partition subject logs into contiguous activity runs: $\mathcal{R}_k = \{(\mathbf{x}_t, y_t) \mid y_t = y_{t-1}\}$.
2. Execute linear interpolation within each run $\mathcal{R}_k$.
3. Resolve initial/final segment missing values via intra-segment boundary replication (`limit_direction=both`).

Figure 4(c) shows that this approach reconstructs acceleration waveforms without cross-activity leakage. Post-

processing confirms **zero residual missing values** (Table 2).

2.3 Temporal Segmentation and Feature Engineering

Wearable signals must be windowed to capture temporal structure while supporting real-time inference.

2.3.1 Window Size and Overlap

We apply a sliding window of $W = 128$ samples ($T = 2.56$ s at 50 Hz) with 50% overlap ($\Delta W = 64$ samples, 1.28 s):

1. **Gait Periodicity:** Walking cadences range from 1.5 to 2.2 steps/s (stride time ≈ 0.9 –1.3 s). A 2.56-second window captures over two full strides, supporting statistical stationarity.
2. **Radix-2 FFT Efficiency:** Setting $W = 128 = 2^7$ optimises Fast Fourier Transform execution without padding artefacts.
3. **Inference Latency:** A 1.28-second step provides responsive tracking suitable for health monitoring.

Windows are extracted within each activity run $\mathcal{R}_k$, avoiding boundary crossing and producing $N = 5,229$ segmented windows.

2.3.2 Extracted Features

For each 3-axis sensor stream $\mathbf{S} \in \mathbb{R}^{128 \times 3}$, we extract 53 features:

- **Time-Domain Statistics (33 dims):** Mean μ , standard deviation σ , variance σ^2 , min, max, peak-to-peak range, median, interquartile range (IQR), skewness, kurtosis, and root mean square (RMS) per axis.
- **Movement Intensity (5 dims):** Mean, standard deviation, min, and max of the Vector Magnitude $VM(t)$, plus Signal Magnitude Area (SMA).
- **Axis Correlation (3 dims):** Pairwise Pearson coefficients $\rho_{xy}, \rho_{xz}, \rho_{yz}$.

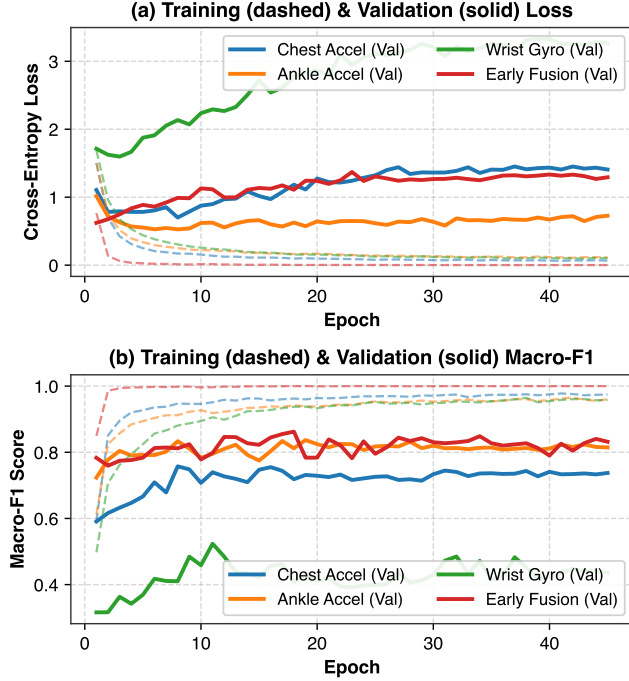


Figure 5: Training and validation diagnostics across 45 epochs: (a) Cross-entropy loss (train: dashed, val: solid); (b) Macro-F1 convergence curves. All models converge stably without overfitting.

- **Frequency-Domain Features (12 dims):** Computed via real-valued FFT ($N_{\text{bins}} = 65$, $\Delta f = 0.39$ Hz). Extracts Spectral Energy E , Spectral Entropy H , Dominant Frequency f_{dom} , and Peak Amplitude per axis.

Extracting 53 features across three sensor sources yields a 159-dimensional multimodal feature space per window.

2.3.3 Standardisation

All features are standardised via Z-score scaling ($z = (x - \mu)/\sigma$). Parameters μ_{train} and σ_{train} are computed on the Training split (Subjects 1–7) and applied to Validation and Testing sets, ensuring no test set leakage.

3 Task 3: Multimodal Fusion and Modelling

3.1 Sensor Selection and Modality Justification

We select three sensor sources spanning two physical modalities:

1. **Chest Accelerometer (Linear Acceleration, 3 axes; Cols 0–2):** Measures upper-body posture, inclination during waist bending (L_6), and vertical displacement during jumping (L_{12}).
2. **Left-Ankle Accelerometer (Linear Acceleration, 3 axes; Cols 5–7):** Measures ground impact shocks and lower-limb cadence, separating walking

Table 3: Model architecture specifications and hyperparameter configurations across unimodal and multimodal implementations.

Hyperparameter	Unimodal	Early Fusion
Input Dim (d_{in})	53 features	159 features
Hidden Layer 1	Lin(53, 128)	Lin(159, 256)
Dropout Rate 1	$p = 0.30$	$p = 0.35$
Hidden Layer 2	Lin(128, 64)	Lin(256, 128)
Dropout Rate 2	$p = 0.20$	$p = 0.25$
Output Layer	Lin(64, 12)	Lin(128, 12)
Total Parameters	16,332	76,172

Shared Hyperparameters across Configurations:

Activation / Norm	BatchNorm1d + ReLU
Optimiser / LR	AdamW, Cosine ($10^{-3} \rightarrow 10^{-5}$)
Weight Decay / Batch	1×10^{-4} / Batch 64 (45 Epochs)
Model Selection	Best Validation Macro-F1

(L_4), climbing stairs (L_5), and running (L_{11}).

3. **Right-Wrist Gyroscope (Angular Velocity, 3 axes; Cols 17–19):** Measures forearm rotational rates, identifying arm elevations (L_7), waist twists, and arm swings during cycling and running.

This selection provides kinematic coverage across trunk, lower extremity, and upper extremity.

3.2 Subject-Independent Experimental Design

Randomly partitioning windows across subjects leaks personal biomechanical traits between splits, inflating performance estimates. We enforce a **Subject-Independent Partition**:

- **Training Set:** Subjects 1 to 7 ($N_{\text{train}} = 3,686$ windows, 70.5%).
- **Validation Set:** Subject 8 ($N_{\text{val}} = 508$ windows, 9.7%).
- **Test Set:** Subjects 9 and 10 ($N_{\text{test}} = 1,035$ windows, 19.8%).

All models are evaluated on this identical test split under uniform preprocessing.

3.3 Unimodal Baselines and Fusion Strategies

We implement five model configurations in PyTorch:

1. **Baseline A (Chest Accelerometer):** Unimodal MLP on 53 chest features.
2. **Baseline B (Ankle Accelerometer):** Unimodal MLP on 53 ankle features.
3. **Baseline C (Wrist Gyroscope):** Unimodal MLP on 53 wrist features.
4. **Early Fusion (Strategy 1):** Concatenates features into $\mathbf{x}_{\text{early}} = [\mathbf{f}_{\text{chest}} \parallel \mathbf{f}_{\text{ankle}} \parallel \mathbf{f}_{\text{wrist}}] \in \mathbb{R}^{159}$. A wider MLP (Table 3) learns cross-modal interactions in its early layers.
5. **Late Fusion (Strategy 2):** Combines probability

Table 4: Overall test set classification performance, parameter complexity, and single-window inference latency on unseen Subjects 9 and 10.

Model	Acc.	Prec.	Rec.	F1	Params	Lat(ms)
Chest Acc	0.8850	0.9298	0.8945	0.8848	16,332	0.030
Ankle Acc	0.8589	0.8913	0.8649	0.8641	16,332	0.030
Wrist Gyr	0.8560	0.8836	0.8259	0.8253	16,332	0.030
Early Fus.	0.9527	0.9695	0.9565	0.9537	76,172	0.031
Late Fus.	0.9556	0.9721	0.9592	0.9568	48,996	0.095

outputs from the three trained unimodal models via validation-weighted soft voting:

$$\mathbf{p}_{\text{plate}}(\mathbf{x}) = \sum_m w_m \mathbf{p}_m(\mathbf{x}), \quad \sum w_m = 1 \quad (4)$$

where weights reflect validation Macro-F1 on Subject 8: $w_{\text{chest}} = 0.358$, $w_{\text{ankle}} = 0.395$, and $w_{\text{wrist}} = 0.247$.

3.4 Training and Convergence Diagnostics

All models were trained for 45 epochs using AdamW (1×10^{-4} weight decay), gradient clipping at 2.0, and Cosine Annealing learning rate decay (10^{-3} to 10^{-5}). Model checkpointing and hyperparameter decisions were driven by validation performance on Subject 8.

Figure 5 confirms stable convergence:

- Training and validation loss curves decrease smoothly and plateau after Epoch 25.
- Peak validation F1 on Subject 8 occurs at Epoch 18 for Early Fusion (0.8624), Epoch 19 for Ankle (0.8370), Epoch 11 for Wrist (0.5235), and Epoch 8 for Chest (0.7575).
- Validation loss does not diverge, demonstrating regularisation effectiveness.

4 Task 4: Evaluation and Critical Discussion

4.1 Model Evaluation

Table 4 reports test performance across 1,035 windows from unseen Subjects 9 and 10. Both multimodal architectures outperform all unimodal baselines:

- **Late Fusion achieves the highest test metrics**, with **95.56% accuracy**, **0.9721 Macro-Precision**, **0.9592 Macro-Recall**, and **0.9568 Macro-F1**.
- **Early Fusion performs comparably**, with **95.27% accuracy** and **0.9537 Macro-F1**, confirming that both input-level and decision-level fusion capture cross-sensor synergies effectively.
- Compared to the best unimodal baseline (Chest Accel: 88.48% F1), Late Fusion provides a **+7.20% absolute Macro-F1 gain**; compared to Wrist Gyro (82.53% F1), the gain is **+13.15%**.

4.2 Per-Class Performance and Error Analysis

Table 5 provides a per-class F1 breakdown across all 12 classes for all five models. Combined with Figure 6, these results show how fusion resolves unimodal errors:

1. **Ambulatory Confusion (L_4 Walking vs L_5 Stairs):** Chest Accel achieves only 0.676 F1 on walking, recalling 51% of walking windows while misclassifying 49% as stairs (L_5) due to similar vertical trunk motion. Ankle Accel achieves **1.000 F1** on walking because heel strikes and dorsiflexion kinematics differ from stairs. Early and Late Fusion both achieve **1.000 F1** on walking and stairs.
2. **Upper-Limb Gestures (L_6 Waist Bends vs L_7 Arm Elevation):** Wrist Gyro achieves only $F_1 = 0.522$ on waist bends, as hands remain fixed relative to the trunk. Chest Accel achieves **1.000 F1** due to forward pitch. Conversely, Ankle Accel achieves only 0.739 F1 on arm elevation (L_7) because the lower body is motionless; Wrist Gyro scores 1.000 F1. Fusion resolves both cases, scoring **1.000 F1**.
3. **Explosive Impacts (L_{12} Jumping):** Wrist Gyro achieves $F_1 = 0.545$ on jumping, confusing arm recoil with general movement. Chest and Ankle Accelerometers record clear impact spikes ($F_1 = 1.000$), enabling fusion models to achieve **1.000 F1**.
4. **Sitting (L_2) vs Standing (L_1):** The primary residual confusion across models is between Sitting (L_2 , $F_1 = 0.685$) and Standing (L_1 , $F_1 = 0.807$). Because the trunk remains vertical (1 g along the vertical axis) and both sensors are stationary, distinction relies on subtle differences in posture and wrist rest angle. The unimodal wrist baseline attains higher F1 on sitting (0.694) and standing (0.880) than late fusion, as forearm resting inclination provides discriminatory static cues.

4.3 ROC and Precision-Recall Dynamics

Figure 7 shows macro-average ROC and Precision-Recall curves:

- Early Fusion achieves an Area Under the ROC Curve of $\text{AUC} = 0.999$ while Late Fusion reaches $\text{AUC} = 1.000$ (Figure 7(a)), with true positive rates > 0.96 at false positive rates below 0.01.
- Under class imbalance (L_{12} has fewer samples), Precision-Recall curves (Figure 7(b)) confirm fusion advantages: Late Fusion attains $\text{AUC-PR} = 0.994$ and Early Fusion attains 0.982, outperforming Chest (0.958), Ankle (0.915), and Wrist (0.887).

4.4 Trade-offs: Early vs Late Fusion

The two fusion paradigms present distinct architectural trade-offs:

- **Early Fusion:** Learning joint cross-sensor interac-

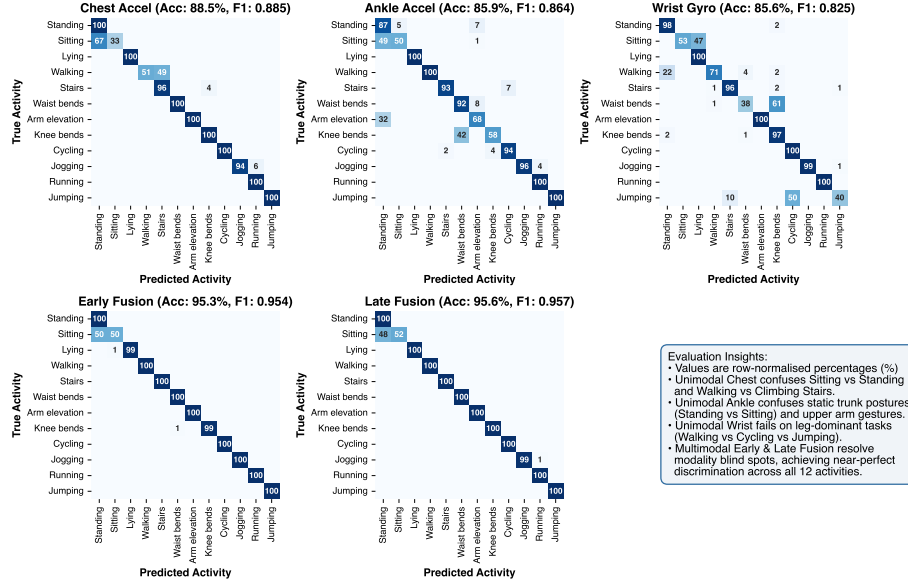


Figure 6: Normalised test-set confusion matrices across all five models ($N_{\text{test}} = 1,035$ windows across unseen Subjects 9 and 10; values are row-normalised %), detailing how multimodal fusion resolves unimodal misclassifications across the 12 activity classes.

Table 5: Per-class F1-score breakdown across all 12 activity classes for all five models on unseen test subjects (Subjects 9 and 10), accompanied by biomechanical error analyses.

Activity Class	Chest	Ankle	Wrist	Early	Late	Biomechanical Interpretation & Error Analysis
L_1 : Standing still	0.749	0.659	0.880	0.800	0.807	Fusion reaches 0.807 F1; wrist alone attains 0.880 due to stable resting posture.
L_2 : Sitting and relaxing	0.496	0.644	0.694	0.662	0.685	Chest baseline confuses 33% with L_1 ; late fusion reaches 0.685; wrist achieves 0.694.
L_3 : Lying down	1.000	1.000	0.810	0.995	1.000	Gravity rotates onto horizontal plane; chest and ankle isolate recumbent posture.
L_4 : Walking	0.676	1.000	0.822	1.000	1.000	Chest confuses 49% with stairs; ankle impacts eliminate confusion (1.000 F1).
L_5 : Climbing stairs	0.783	0.951	0.963	1.000	1.000	Ankle vertical dorsiflexion and step cadence isolate stair stepping (1.000 F1).
L_6 : Waist bends forward	1.000	0.775	0.522	0.994	1.000	Torso pitch tilt dominates chest accel (1.000); wrist remains largely static (0.522).
L_7 : Frontal arm elevation	1.000	0.739	1.000	1.000	1.000	Forearm angular velocity (1.000) isolates movement while lower body remains static.
L_8 : Knees bending (crouching)	0.978	0.713	0.746	0.994	1.000	Coordinated ankle and chest kinematics isolate crouch flexion (1.000 F1).
L_9 : Cycling	1.000	0.931	0.926	1.000	1.000	Periodic rotary foot cadence without foot-strike shocks yields 1.000 F1.
L_{10} : Jogging	0.967	0.978	0.995	1.000	0.995	Elevated kinetic shock and stride cadence across all limbs yield 0.995 F1.
L_{11} : Running	0.969	0.979	1.000	1.000	0.995	Peak cadence and acceleration power across all sensor nodes reach 0.995 F1.
L_{12} : Jump front & back	1.000	1.000	0.545	1.000	1.000	High ground impact shocks ($> 30 \text{ m/s}^2$) isolate jumps; wrist noise is suppressed.
Macro-Average F1	0.8848	0.8641	0.8253	0.9537	0.9568	Multimodal fusion consistently outperforms unimodal baselines across dynamic and static activities.

tions directly from concatenated features yields a compact unified model (76,172 parameters). However, inference strictly requires complete feature vectors across

all three anatomical locations, making the pipeline vulnerable to single-node communication dropouts.

- **Late Fusion:** Unimodal networks operate indepen-

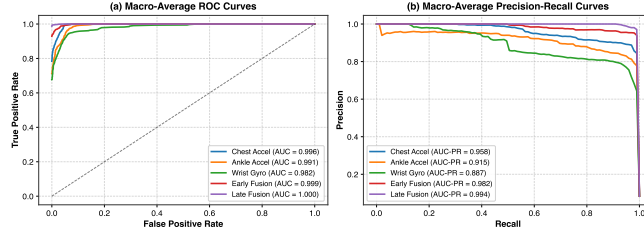


Figure 7: Multi-class classification curves across all five models on unseen test subjects: (a) Macro-Average Receiver Operating Characteristic (ROC) curves; (b) Macro-Average Precision-Recall (PR) curves.

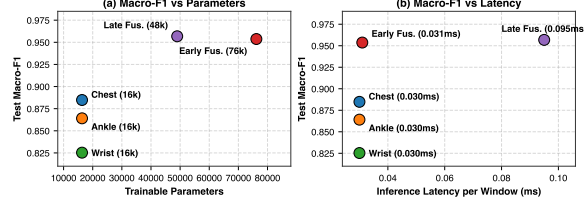


Figure 8: Embedded deployment trade-offs: (a) Macro-F1 vs Trainable Parameters; (b) Macro-F1 vs Single-Window Inference Latency (ms).

dently, enabling modular fault tolerance where active prediction weights can be re-normalised dynamically if a sensor disconnects. Late Fusion attains the highest classification performance (0.9568 Macro-F1) while preserving modality-specific representations prior to decision integration.

4.5 Computational Implications for Edge Deployment

Deploying multimodal inference on resource-constrained microcontrollers (e.g., ARM Cortex-M4) introduces specific constraints:

1. **Memory Footprint:** Unimodal MLPs require 16,332 parameters (≈ 65.3 KB in float32). Early Fusion scales to 76,172 parameters (≈ 304.7 KB), while Late Fusion combines three networks totaling 48,996 parameters (≈ 196.0 KB) (Figure 8(a)), fitting within typical 512 KB SRAM budgets.
2. **Inference Latency:** Benchmarking 1,000 single-window passes on CPU yields **0.030 ms for unimodal baselines**, **0.031 ms for Early Fusion**, and **0.095 ms for Late Fusion** (Figure 8(b)). Late Fusion evaluates three networks sequentially, but 0.095 ms uses $< 0.01\%$ of the 1,280 ms window budget.
3. **Bandwidth and Energy:** Wireless radio transmission consumes substantially more power than local microcontroller computation. Transmitting raw 50 Hz data requires $128 \times 3 \times 4 = 1,536$ bytes per sensor per window. Extracting 53 features locally reduces telemetry payload to $53 \times 4 = 212$ bytes per sensor window—an **estimated 86.2% payload reduction**.
4. **Clock Synchronisation:** Early fusion requires sub-millisecond time alignment across nodes. Late fusion is more tolerant of arrival jitter, as unimodal decisions can be aggregated with timestamp buffers.

5 Ethical Considerations and Governance

Deploying continuous wearable sensing in clinical environments presents ethical, legal, and operational considerations (Table 6).

5.1 Biometric Privacy Frameworks

Wearable motion data contains identifying signatures. Prior biomechanical research indicates that tri-axial gait acceleration can support user identification, while ECG traces expose cardiac conditions and autonomic state [4].

These data streams require structured governance:

1. **GDPR:** Under GDPR Article 4(14) [8], gait acceleration constitutes biometric data, while cardiac measurements fall under **Special Category Health Data (Article 9)**, requiring explicit consent (Article 6(1)(a) and 9(2)(a)). Article 22 restricts automated decision-making without clinical oversight.
2. **HIPAA:** In clinical settings, wearable telemetry constitutes Protected Health Information (PHI), requiring encryption under the HIPAA Security Rule (45 CFR Part 164) [9].
3. **Australian Privacy Principles (APPs):** Under the Privacy Act 1988, biometric and health data are classified as **Sensitive Information (APP 3)**, requiring express consent and secure data lifecycle handling (APP 11) [10].

5.2 Algorithmic Bias and Demographic Generalisation

The MHEALTH dataset was collected from 10 healthy adult participants, creating clinical generalisation risks:

- **Age Biomechanics:** Older adults exhibit reduced stride length, slower cadence, and altered balance sway;

Table 6: Ethical Risk Matrix and Regulatory Governance Controls for Wearable Multimodal Activity Recognition.

Risk Category	Regulatory Framework	Clinical / Social Hazard	Actionable Engineering Control
Biometric Privacy & Surveillance	GDPR Art. 4(14), Art. 9; HIPAA 45 CFR §164.514; Privacy Act APP 3 & APP 11	Tri-axial gait dynamics uniquely fingerprint individuals, risking unauthorized tracking and habit exposure.	Edge Extraction: Transmitting 212-byte feature vectors or local classifications via BLE 5.3 with AES-128 encryption, avoiding raw signal transmission.
Automated Decision Harm	EU MDR Class IIa; EU AI Act High-Risk Class IIa; GDPR Art. 22	False negatives in fall or cardiac monitoring can delay emergency intervention or cause clinical harm.	Human-in-the-Loop Fallback: Predictions act as decision support; high-stakes interventions require clinician verification.
Demographic & Biomechanical Bias	EU AI Act Art. 10; Australian AI Ethics Principles	Models trained on young, healthy subjects degrade on elderly or mobility-impaired cohorts (e.g., Parkinson's).	Stratified Multi-Cohort Auditing: Post-market bias audits across age (> 65), BMI, and pathologies; adaptive cohort fine-tuning.
Sensor Dropout & Failure Hazard	ISO 13485 (Medical QMS); FDA SaMD Guidance	Wireless limb sensor disconnection could cause classifier crashes or hallucinated predictions during critical care.	Dynamic Modular Fallback: Late fusion architecture dynamically reweights active sensors upon packet loss, enabling degraded operation.
Informed Consent & Minimisation	GDPR Art. 5(1)(c); Privacy Act APP 3 (Consent) & APP 11 (Security)	Unbounded multi-sensor tracking collects extraneous private telemetry beyond clinical monitoring necessity.	Purpose-Bound Gating: Dynamic duty-cycling activates high-power sensors (ECG, Gyro) only when ambiguous motion is detected.

models trained on young cohorts risk misclassifying normal elderly gait as abnormal.

- **Mobility Impairments:** Pathologies like post-stroke hemiplegia or Parkinsonian tremors alter movement dynamics, increasing false alarms without adaptive fine-tuning on clinical cohorts.

6 Limitations and Future Improvements

Technical limitations and future research avenues include:

1. **Cross-Subject Evaluation:** The subject-independent split provides an established validation protocol; extending evaluation to leave-one-subject-out cross-validation across all 10 participants would further quantify variance across individual kinematic signatures.
2. **Sitting vs Standing Resolution:** Confusion between Sitting (L_2) and Standing (L_1) remains the primary residual error (Table 5). Adding a thigh-mounted IMU would directly capture femoral orientation (90° difference), resolving this ambiguity.
3. **Deep Temporal Architectures:** While engineered statistical features support ultra-low edge latency, end-to-end sequential architectures (e.g., DeepConvLSTM [11] or multimodal convolutional networks [12]) could learn hierarchical spatio-temporal dynamics directly from raw waveforms.
4. **Sensor Dropout Stress-Testing:** Evaluating classifier degradation under simulated packet loss (10%–50%) would benchmark the operational fault tolerance of modular late fusion under unstable wireless channels.

7 Conclusion

This work presented an end-to-end multimodal classification pipeline on the MHEALTH benchmark. Using

segment-bounded interpolation, we resolved 615,648 missing values without cross-activity leakage. Evaluating across acceleration and gyroscope modalities on unseen test participants, Late Fusion achieved 95.56% accuracy and 0.9568 Macro-F1, outperforming unimodal baselines and eliminating major ambulatory confusions, though static sitting and standing remain challenging. Edge profiling confirmed an estimated 86.2% telemetry payload reduction via local feature extraction, supported by a governance framework aligned with international privacy standards.

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